

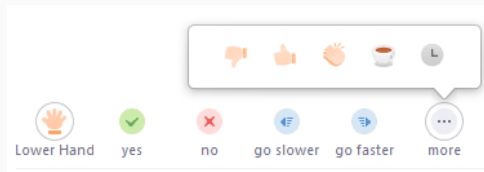
Algorithms for Lattice Fermions User Workshop 2020

5-9th Oct 2020

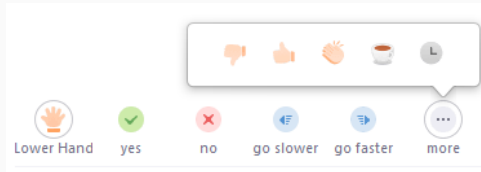
Welcome



1. Please **raise** your hands:



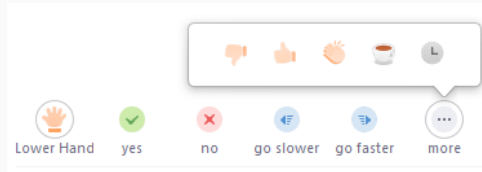
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2. Keep yourself **muted** during presentations in order to avoid background noise.

Organizational remarks

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2. Keep yourself **muted** during presentations in order to avoid background noise.
3. Got any questions/comments? Please use the **chat**.
4. If you encounter any issues, please let us know in the chat box or send a mail to alf@physik.uni-wuerzburg.de

1. Organizational remarks

2. The ALF Project

ALF Features

ALF Team

ALF Timeline & Sponsors

3. Workshop Program

Algorithms for Lattice Fermions: general code for auxiliary-field quantum Monte Carlo simulations

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General Hamiltonian

- single-body operators,
- squares of single-body op.
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Advanced Features

- parallel tempering
- ground state projective QMC
- global Monte Carlo updates
- continuous fields
- Langevin dynamics updates
- stochastic Maximum Entropy method
- symmetric Trotter decomposition

ALF Team

ALF Team members:



Fakher Assaad



Florian Goth



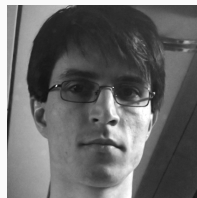
Martin Bercx



Johannes Hoffmann



Jonas Schwab



Jefferson S.E. Portela

A short history of ALF:

2016 ALF 0.5

2017 ALF 1.0

Martin Bercx, Florian Goth, Johannes S. Hofmann, and F.F.A, *The ALF (Algorithms for Lattice Fermions) project release 1.0. Documentation for the auxiliary field quantum Monte Carlo code*, SciPost Phys. **3** (2017), 013.

2020 ALF 2.0, pyALF

preprint: [Documentation/doc.pdf](#)



SFB1170
ToCoTronics



Leibniz-Rechenzentrum
der Bayerischen Akademie der Wissenschaften



GCS
Gauss Centre for Supercomputing



Center of excellence – complexity and
topology in quantum matter



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- **Contributed talks** – Participants' presentations:
 - Wed. 14:00 – Miklós Werner, *Matrix product state simulations with general non-Abelian symmetries*
 - Wed. 14:30 – Joel Hutchinson, *Fractional topology from entanglement*
 - Thu. 10:00 – Tarun Grover, *Simulatable models of non-Fermi liquids and nodal superconductivity*
 - Thu. 14:00 – Izabella Lovas, *Quantum Coulomb glass on the Bethe lattice*
 - Thu. 14:30 – Natanael Costa, *Phase diagram of the two-dimensional Hubbard-Holstein model*

Workshop Program

	Thursday 01/10 (GMT+2)	Monday 05/10	Tuesday 06/10	Wednesday 07/10	Thursday 08/10	Friday 09/10
10:00	ALF / pyALF Installation		Tutorial	Tutorial	Contributed Talk	Workgroup
11:00					Workgroup	
12:00			Tutorial	Tutorial	Workgroup	Workgroup
13:00			ALF basics: <i>Observ.</i>	ALF basics: <i>Hopp.</i>		
14:00		Welcome	Break Room	Break Room		
15:00		Kick-off Talk <i>Fermion Monte Carlo</i> <i>F.F. Assaad</i>	<i>Fermion Monte Carlo</i> + Project Proposals	Contributed Talks	Contributed Talks	Workgroup Presentations
16:00		Break Room	Break Room	Break Room	Break Room	
17:00		Tutorial: <i>Intro.</i>	Tutorial: <i>Advanced</i>	<i>Fermion Monte Carlo</i> + Project Proposals (2)	Discussion	
18:00	ALF / pyALF Installation (GMT+2)	ALF basics: <i>Lattices</i>	ALF basics: <i>Interact.</i>	ALF basics: <i>Models</i>	Workgroup	Discussion
		Tutorial	Tutorial	Workgroup		Break Room

Pre-Conference

Getting to Know ALF

ALF in Production

Key:

Plenary/synchronous

Break-out rooms/on-call

ALF *basics*